Mastering Precision, Recall, and F1 Scores for Skewed Datasets

A structured tutorial on evaluation metrics for imbalanced machine learning problems

From the failure of raw accuracy to the Confusion Matrix, thresholding, and the F1 Score — with worked numerical examples.



10 sections • colour-coded by topic

1 | Full Walkthrough: Navigating the Lecture

This session provides a structured journey from the initial failure of standard intuition to the implementation of robust engineering solutions for evaluating machine learning models. We begin by examining why traditional accuracy metrics collapse when applied to skewed datasets, where the target class is rare. To move beyond these deceptive figures, we introduce the Confusion Matrix, a tool used to categorise prediction outcomes into four distinct quadrants: True Positives, True Negatives, False Positives, and False Negatives.

Building upon this matrix, we define the precision-recall framework as a more nuanced way to evaluate performance. We then explore the "Engineering Philosophy" of thresholding, demonstrating how we can manually adjust model behaviour to reflect real-world costs and benefits. Finally, the lecture synthesises these concepts into the F1 Score, an automated metric that allows us to compare different learning algorithms efficiently. Looking ahead, mastering these evaluative tools provides the necessary foundation for our next major topics: building complex systems using Neural Networks and Decision Trees.

2 | Core Concept: The Intuitive Foundation

Before diving into the mathematics, it is essential to grasp the fundamental problem: when "positive" results are rare, simple accuracy is a deceptive metric. In skewed datasets, a model can appear highly successful simply by predicting the most common outcome every time. To develop a truly useful diagnostic tool, we must adopt metrics that specifically measure the model's ability to identify the rare class accurately without producing an excessive number of false alarms.

3 | Why It Matters: The Accuracy Paradox

To understand why we must move beyond accuracy, let us consider the "Rare Disease" scenario found in clinical diagnostics.

- **The Problem with Accuracy:** Imagine a population where only 0.5% of patients have a specific disease. A student might be pleased with an algorithm that achieves 99% accuracy (1% error). However, a "dumb," non-learning algorithm that simply predicts y = 0 for every single patient would achieve 99.5% accuracy (0.5% error).
- **The Distinguishability Crisis:** In this context, accuracy fails to help us distinguish which model is actually superior. We cannot easily tell if an algorithm with 99.2% accuracy is "better" than one with 99.5% if the lower-accuracy model actually identifies patients with the disease, while the higher-accuracy model identifies none at all.
- **The Goal:** Our objective is to create a "useful diagnostic tool" rather than just chasing a high-accuracy number that masks a complete failure to detect the rare class.

4 | The Mechanism: From Matrix to Metrics

To evaluate a model's performance on skewed data, we construct a Confusion Matrix to track the intersection of the actual labels and the algorithm's predictions.

The Confusion Matrix

	Actual Class: 1	Actual Class: 0
Predicted Class: 1	True Positive (TP)	False Positive (FP)
Predicted Class: 0	False Negative (FN)	True Negative (TN)

- **True Positive (TP):** The algorithm predicted 1, and the actual label was 1.
- **True Negative (TN):** The algorithm predicted 0, and the actual label was 0.
- **False Positive (FP):** The algorithm predicted 1, but the actual label was 0.
- **False Negative (FN):** The algorithm predicted 0, but the actual label was 1.

Mnemonic: the first word (True/False) indicates if the prediction was correct; the second word (Positive/Negative) indicates the algorithm's prediction.

Key Metrics

Precision (P) measures what fraction of patients predicted to have the disease actually possess it — the fraction of predicted positives that were correct.

FORMULA

```
Precision = TP / (TP + FP) # fraction of predicted positives that were correct
```

Recall (R) measures what fraction of all actual cases of the disease were successfully detected by the algorithm.

FORMULA

```
Recall = TP / (TP + FN) # fraction of actual positives that were detected
```

Boundary Cases

If an algorithm predicts y = 0 for every example, the Recall is 0 because the numerator (TP) is 0. In this scenario, Precision is technically undefined because the denominator (TP + FP) is 0, resulting in a division by zero; however, in practice, we treat such cases as 0.

5 | Threshold Trade-offs: Adjusting the Philosophy

Having established our metrics, we can now look at how to optimise the performance of our learning algorithm by adjusting the classification threshold. In Logistic Regression, we typically predict y = 1 if f(x) ≥ 0.5. By shifting this threshold, we can change the model's "philosophy."

- **High Threshold (e.g., 0.7 or 0.9):** We predict y = 1 only if we are very confident. This is common when the treatment is invasive or expensive. *Result: Higher Precision, Lower Recall.*
- **Low Threshold (e.g., 0.3):** We predict y = 1 whenever there is a reasonable doubt, aiming to be safe. This is used when leaving a disease untreated has fatal consequences but the treatment is relatively safe. *Result: Lower Precision, Higher Recall.*

The Precision-Recall Curve

By plotting these different points (e.g., from a threshold of 0.99 to 0.01), we create a curve. Selecting the best point on this curve is a manual decision based on balancing the cost of False Positives against False Negatives.

6 | The F1 Score: Automated Synthesis

To compare algorithms effectively without manually inspecting two different metrics, we require a single evaluation score.

The Failure of Arithmetic Mean

Taking a simple average of P and R is insufficient. Consider Algorithm 3 (detailed below), which "cheats" by predicting y = 1 for every patient. This results in a perfect Recall of 1.0 but a near-zero Precision. An arithmetic mean would still produce a high score (> 0.5), making a useless algorithm look superior.

The F1 Formula

$$F_1 = 2\frac{PR}{P+R}$$

FORMULA

```
F1 = 2 × (P × R) / (P + R)
# harmonic mean of Precision and Recall
F1 = 1 / ( 0.5 × (1/P + 1/R) )
```

$$F_1 = \frac{2}{\left(\frac{1}{P} + \frac{1}{R}\right)}$$

Intuition: The F1 Score gives greater emphasis to whichever value is lower. If either Precision or Recall is very small, the F1 Score will be low, ensuring that only models with a healthy balance of both metrics achieve a high rating.

7 | Worked Examples: The Numerical Evidence

Example 1: Matrix Calculations

Using 100 cross-validation examples:

- TP: 15
- FP: 5
- FN: 10
- TN: 70

CALCULATION

```
Precision = 15 / (15 + 5) = 0.75
Recall    = 15 / (15 + 10) = 0.60
```

Example 2: Algorithm Comparison

Algorithm	Precision (P)	Recall (R)	Average (Mean)	F1 Score
Algorithm 1	0.5	0.4	0.45	0.444
Algorithm 2	0.7	0.1	0.40	0.175
Algorithm 3	0.02	1.0	0.51	0.0392

In this table, Algorithm 3 demonstrates how the F1 Score identifies “cheating” models that have high recall simply by predicting positive for everyone.

8 | Foundational vs. Incidental Concepts

To assist your studies, concepts are categorised here by their instructional priority.

Foundational (Master These)

- The 2×2 Confusion Matrix
- The mathematical definitions of Precision and Recall
- The impact of thresholding on the trade-off
- The logic of the F1 Score

Incidental (Look Up Later)

- The formal name “Harmonic Mean”
- Specific arbitrary threshold values (like 0.7)
- The specific context of the rare disease once the logic of skewed data is understood

9 | Real-World Relevance: Engineering Scenarios

The decision of where to set your threshold depends entirely on the real-world consequences of your errors.

- **High Precision Priority:** Essential when the action taken is invasive, painful, or expensive. You only want to predict y = 1 when you are highly certain, to avoid unnecessary patient trauma or cost.
- **High Recall Priority:** Necessary when the consequences of a False Negative (leaving a condition untreated) are fatal. In these cases, it is better to be “safe” and predict y = 1 even with lower confidence.

10 | Common Misconceptions and FAQs

Misconceptions

- “High accuracy always means a good model”: As we have seen, in skewed data, a 99.5% accurate model can be completely useless if it never detects the rare class.
- “Precision and Recall always move in the same direction”: There is almost always a trade-off; as you increase the threshold to improve Precision, your Recall will inevitably drop.

FAQs

1. **Why not use the average of P and R?** The arithmetic mean does not penalise models with a near-zero value in one metric. The F1 Score ensures that if either metric is poor, the overall score remains low.
2. **What happens to Precision if the threshold is 0.99?** Precision will become very high because you are only predicting positive when you are "very confident." However, your Recall will likely drop to a very low level.
3. **How do you handle a model that never predicts a positive case?** The Recall is 0. While the Precision formula would result in a division by zero, we set it to 0 in practice. Any model with an F1 Score of 0 is considered not useful.

Takeaway: Raw accuracy is misleading on skewed datasets. Use the Confusion Matrix to compute Precision and Recall, tune the decision threshold to match the real-world cost of false positives versus false negatives, and use the F1 Score — the harmonic mean of P and R — whenever you need one number to fairly compare algorithms, since it punishes any model that is lopsided in either metric.